Notes Week 12

a. Analytic

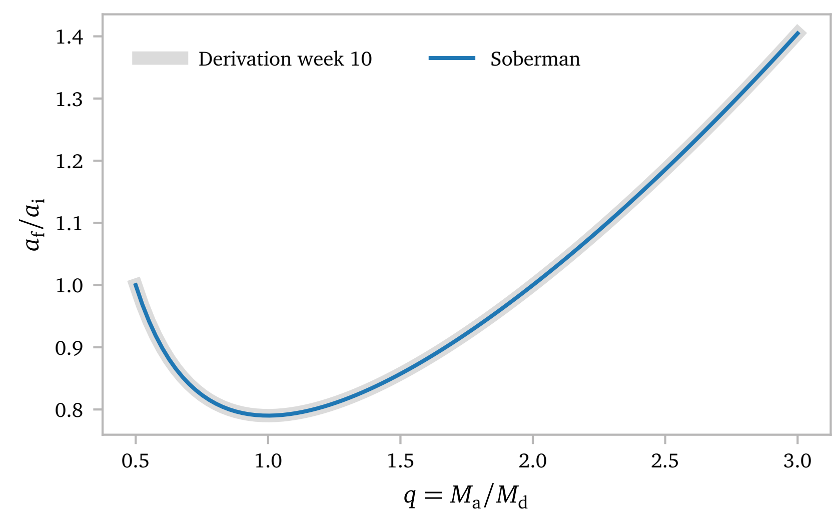
I am basing my implementation on the results from Soberman et al. (1997).

The code below shows my implementation.

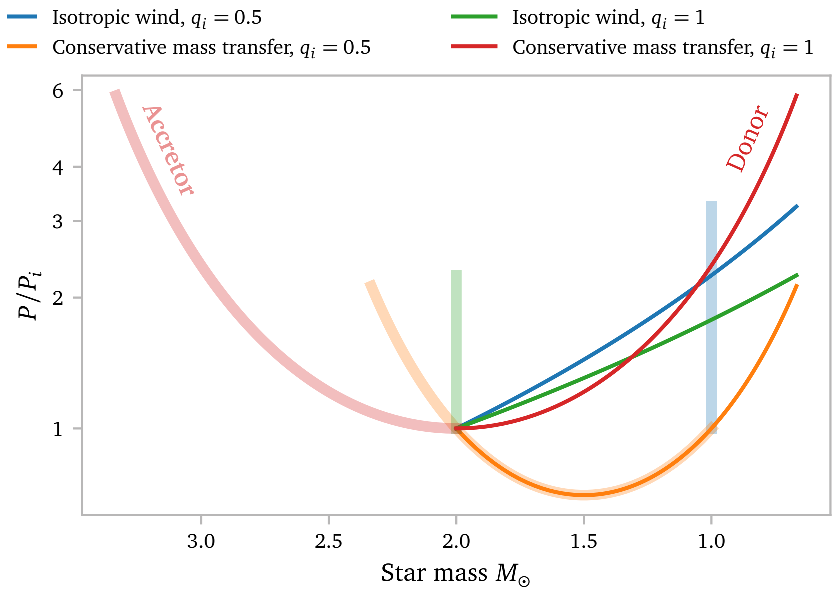
```
class CombineEvolve:
    def __init__(self, alpha, beta, delta, gamma, qs, A=1, M_1_i=2):
        self.alpha = alpha  # fraction lost from donor as wind
        self.beta = beta  # fraction lost from accretor as wind
        self.delta = delta  # fraction transferred to CCT
        self.gamma = gamma  # measure for CCT distance
        self.epsilon = 1 - self.alpha - self.beta - self.delta
        self.qs = 1 / qs # note that soberman1997 uses q = m_d / m_a
        self.q0 = 1 / qs[0]

        self.A = A  # enhancement factor
        self.M_1_i = M_1_i  # initial mass of donor

        self.curA = self._comp_curA()
        self.curB = self._comp_curB()
        self.curC = self._comp_curC()

    def _comp_curA(self):
        # equation B8 soberman1997
        return self.A * self.alpha + self.gamma * self.delta

    def _comp_curB(self):
        # equation B9 soberman1997
        return (self.A * self.alpha + self.beta) / (1 - self.epsilon)

    def _comp_curC(self):
        # equation B10 soberman1997
        term_1 = (self.gamma * self.delta * (1 - self.epsilon)) / self.epsilon
        term_2 = (self.A * self.alpha * self.epsilon) / (1 - self.epsilon)
        term_3 = (self.beta) / (self.epsilon * (1 - self.epsilon))
        return term_1 + term_2 + term_3

    @property
    def a_over_a0(self):
        # table 5 soberman1997
        term_1 = (self.qs / self.q0) ** (2 * self.curA - 2)
        term_2 = ((1 + self.qs) / (1 + self.q0)) ** (1 - 2 * self.curB)
        term_3 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (
            2 * self.curC + 3
        )
        return term_1 * term_2 * term_3

    @property
    def M_tot_over_M_tot0(self):
        # table 5 soberman1997
        term_1 = (1 + self.qs) / (1 + self.q0)
        term_2 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (-1)
        return term_1 * term_2

    @property
    def P_over_P0(self):
        # table 5 soberman1997
        term_1 = (self.qs / self.q0) ** (3 * self.curA - 3)
        term_2 = ((1 + self.qs) / (1 + self.q0)) ** (1 - 3 * self.curB)
        term_3 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (
            3 * self.curC + 5
        )
        return term_1 * term_2 * term_3

    @property
    def M_tot(self):
        return self.M_1_i * (1 + 1 / self.q0) * self.M_tot_over_M_tot0

    @property
    def M_d(self):
        return self.M_tot / (1 + 1 / self.qs)

    @property
    def M_a(self):
        return 1 / self.qs * self.M_d
```

First, I want to compare my implementation of the equations from Soberman et al. (1997) with my results from week 10, which are specifically derived for the case of conservative mass transfer. This aligns with $\varepsilon = 1$ and α, β and $\delta = 0$.

Plugging these values into my Soberman et al. (1997) implementation for a given starting q, we arrive at the following plot:



The curves align, and show the expected behaviour. The ratio of a_f/a_i has a minimum at $q = 1$.



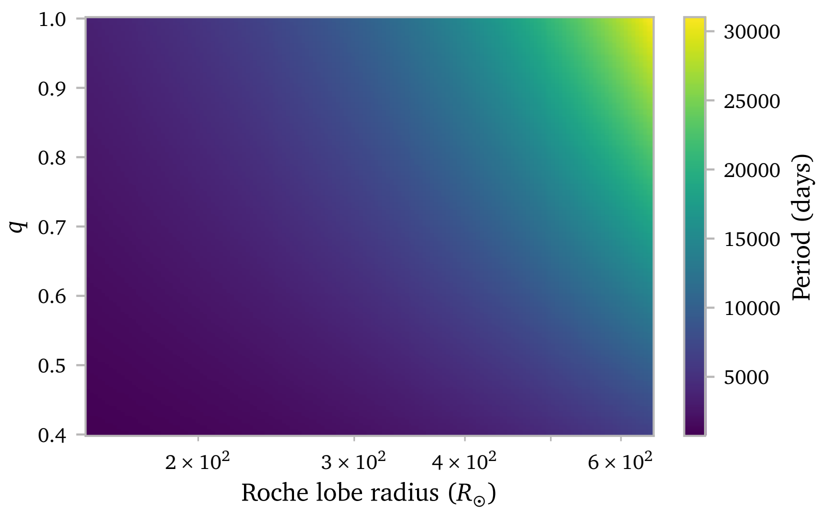
The plot shows four different evolutions. The green and blue lines show a binary system where mass is lost via isotropic winds, while the orange and red lines show a binary system where mass is transferred.

In the case of conservative mass transfer, the lines for the donor and accretor are symmetric, which makes sense. In the case of isotropic winds, the accretor does not change, which also makes sense.

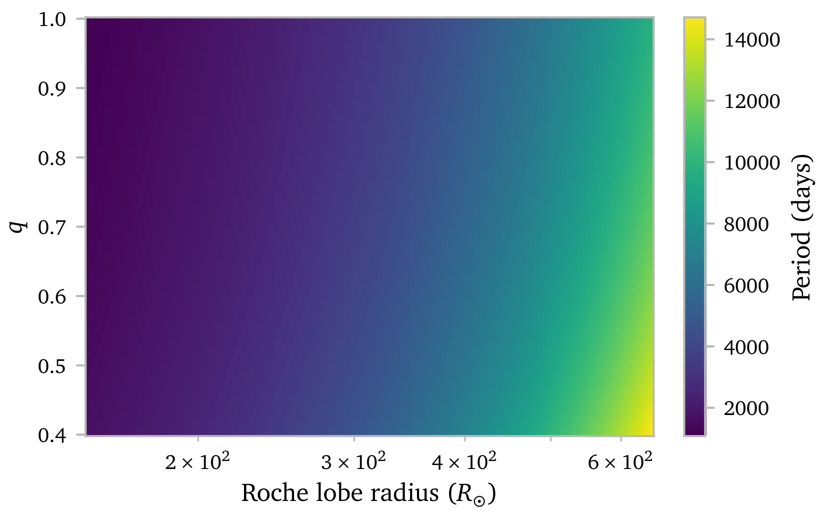
The effect of the mass ratio $q \equiv M_a/M_d$ is inverted for the two mass loss mechanisms.

We can analytically compute a grid of models, similar to the grid of models that we simulate using MESA.

a.1 Grids

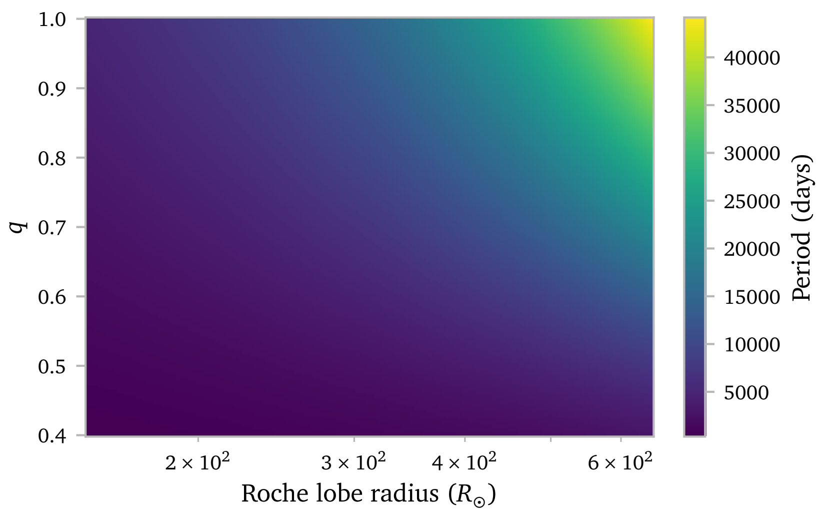


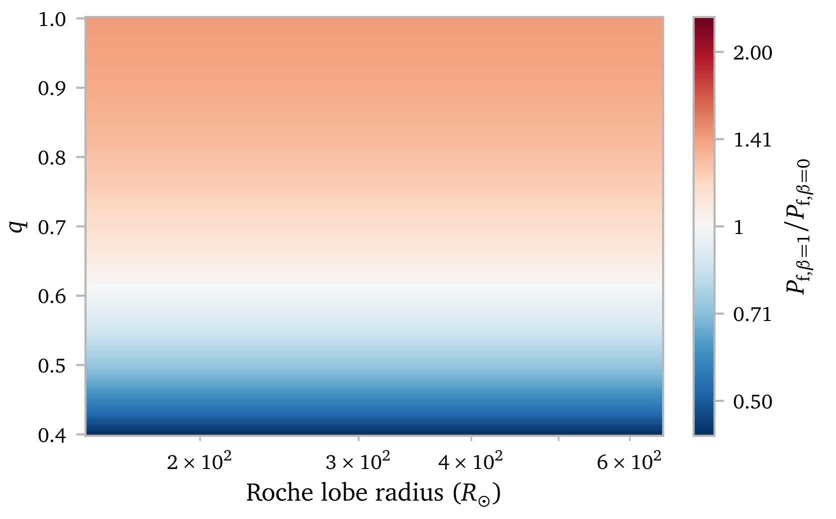
For example, here I show the final orbital period for models that undergo exclusively conservative mass transfer ($\alpha, \beta, \delta = 0$). I make sure the evolution proceeds until the donor star has a mass of 0.6 $M_\odot$, which is just the core. I think the results make perfect sense.



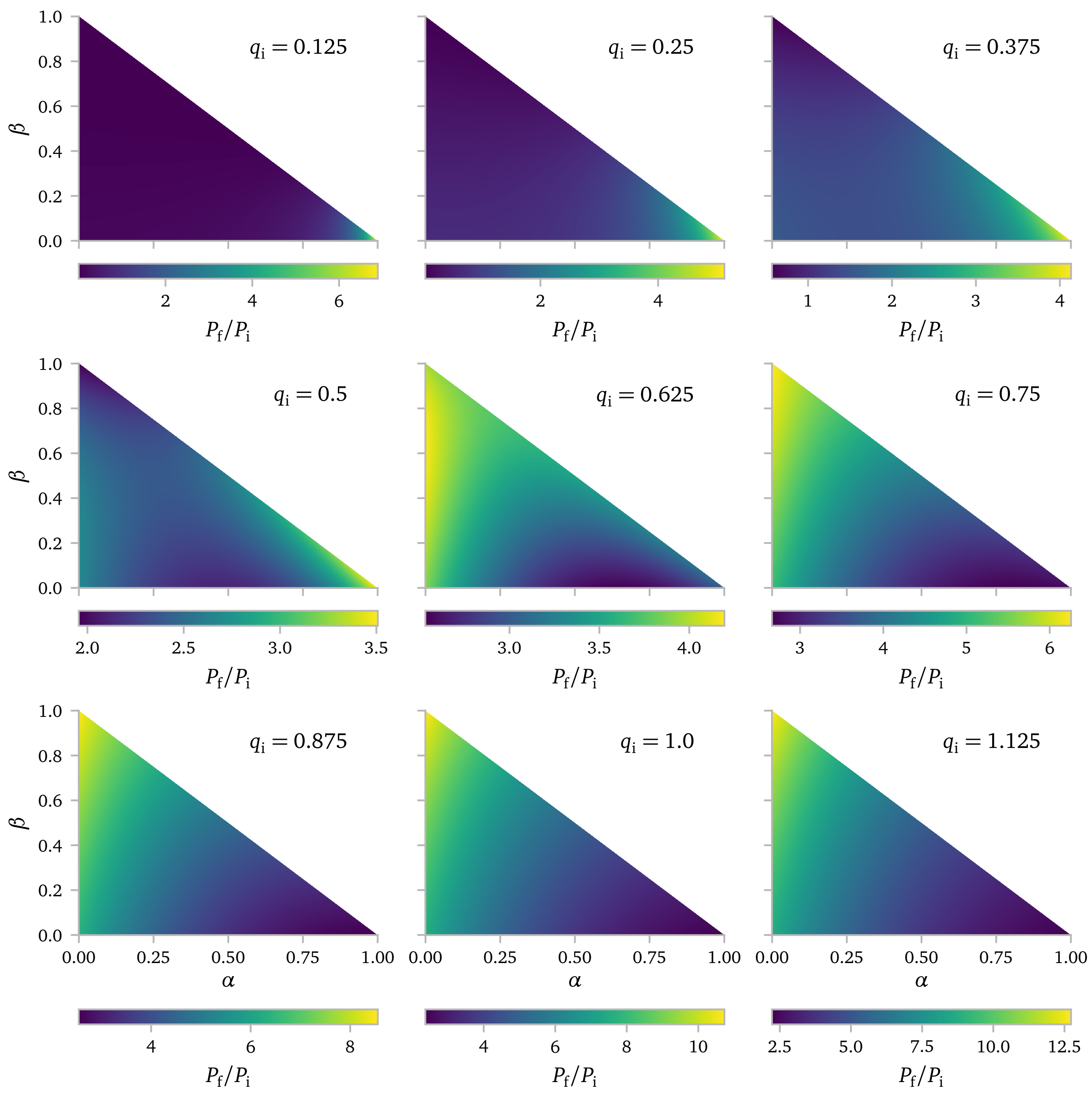
Similarly, we can plot the final orbital periods in the case of only wind mass loss and no accretion, corresponding to $\alpha = 1, \beta = \delta = 0$.

Again, I think these results make sense. We have already seen that wind mass loss has an inverse q dependence on the initial to final period, with lower q generating a greater $\frac{P_f}{P_i}$. Logically, the largest periods are now found for low q and high initial R_{RL}. Overall the periods are smaller, which is also expected.



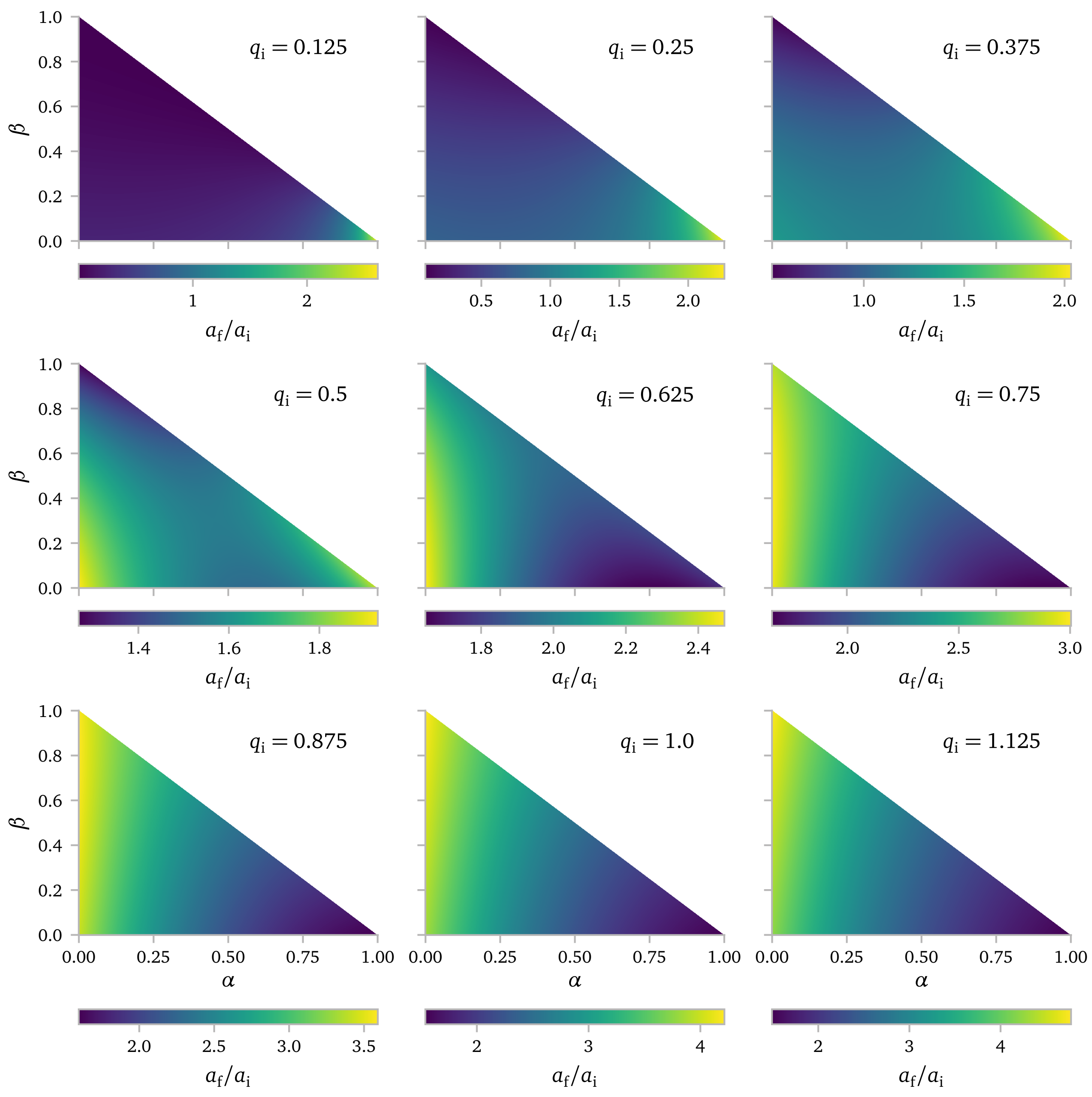


We can also look at the effects of α and β on the orbits. The first plot here shows how period changes with respect to the initial period. I made 9 different plots for increasing initial mass ratios. Every point corresponds with a binary system where the donor initially has a mass of 2 $M_\odot$ and ends up with a mass of 0.6 $M_\odot$. The plots are in the shape of triangles as the following condition *needs* to be true: $\alpha + \beta < 1$.



As can be seen, the initial mass ratio quite significantly alters the effects of α and β. For relatively low initial mass ratios, the orbits will shrink in response to very high β and grow in response to very high α. The inverse is true for mass ratios close to unity.

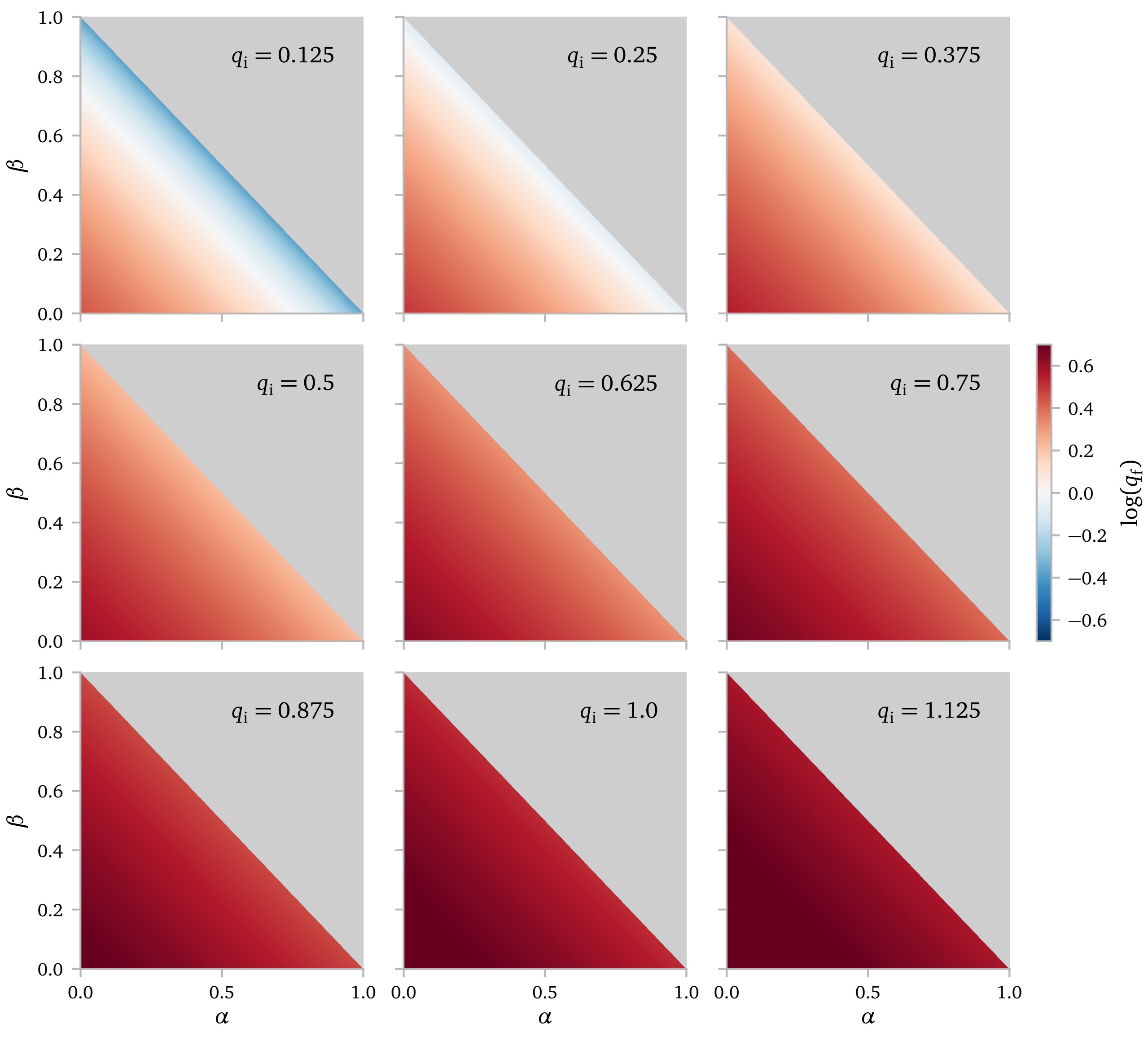
A very similar picture arises when we look at the final separation over the initial separation.



The qualitative behaviour is the same, but the actual differences in ratios are much smaller. Interestingly as well, for low q, the orbits can actually shrink with respect to the initial orbital separation.

These results were not really what I expected. I thought the orbits would *always* shrink when a significant fraction of the mass is lost when accreted onto the accretor. However, this does not seem to be the case. To get a better intuition for the cases where the orbit does shrink with respect to the $\beta = 0$ case, I plot the final and average mass ratios of the binary evolution below. These plots are very structured, which makes sense. The plots also *share* a diverging colorscale. The divergence point lies at $\log q = 0$, which aligns with the case where the stars are equally heavy.

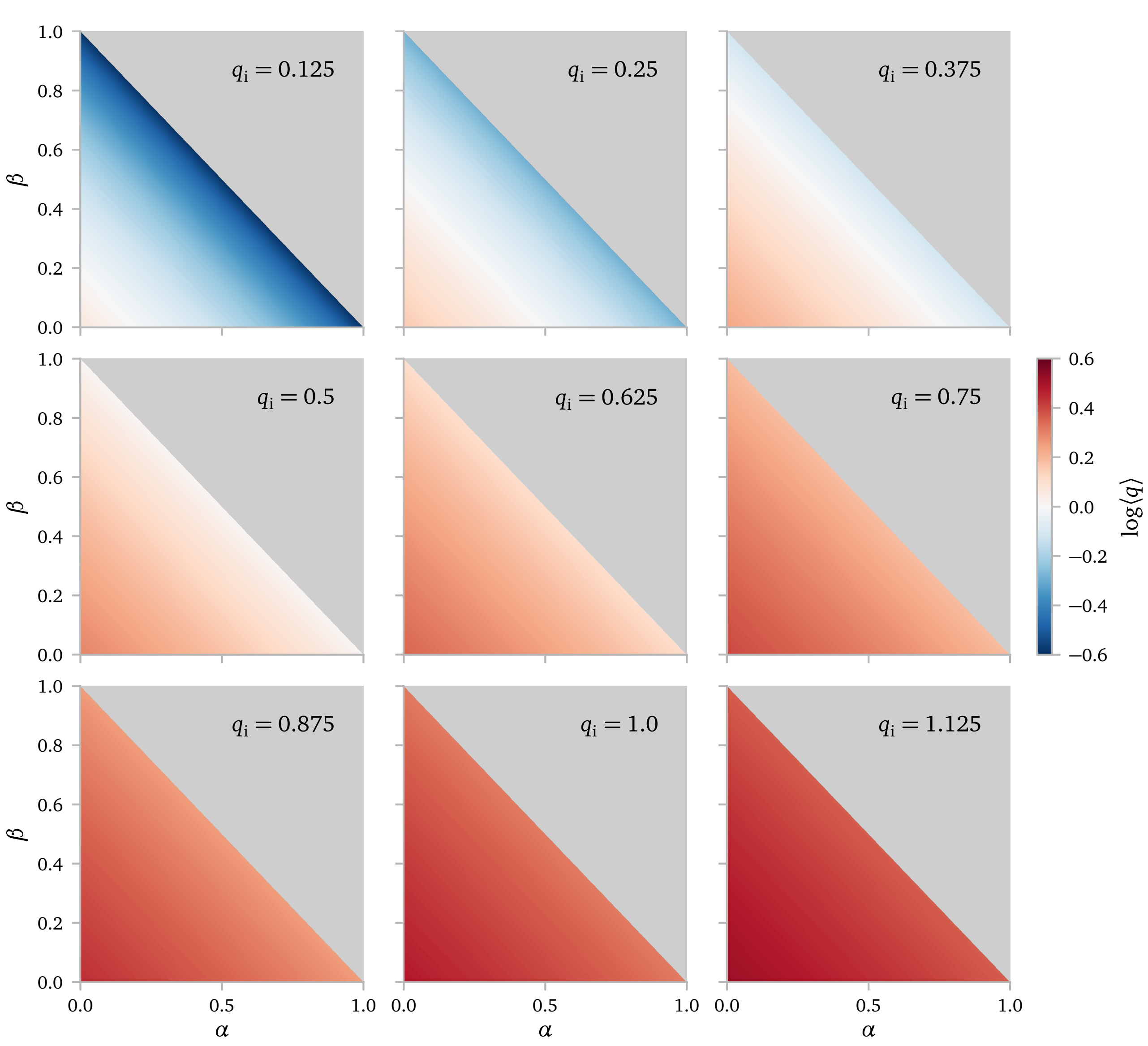
I start by plotting the final mass ratios.



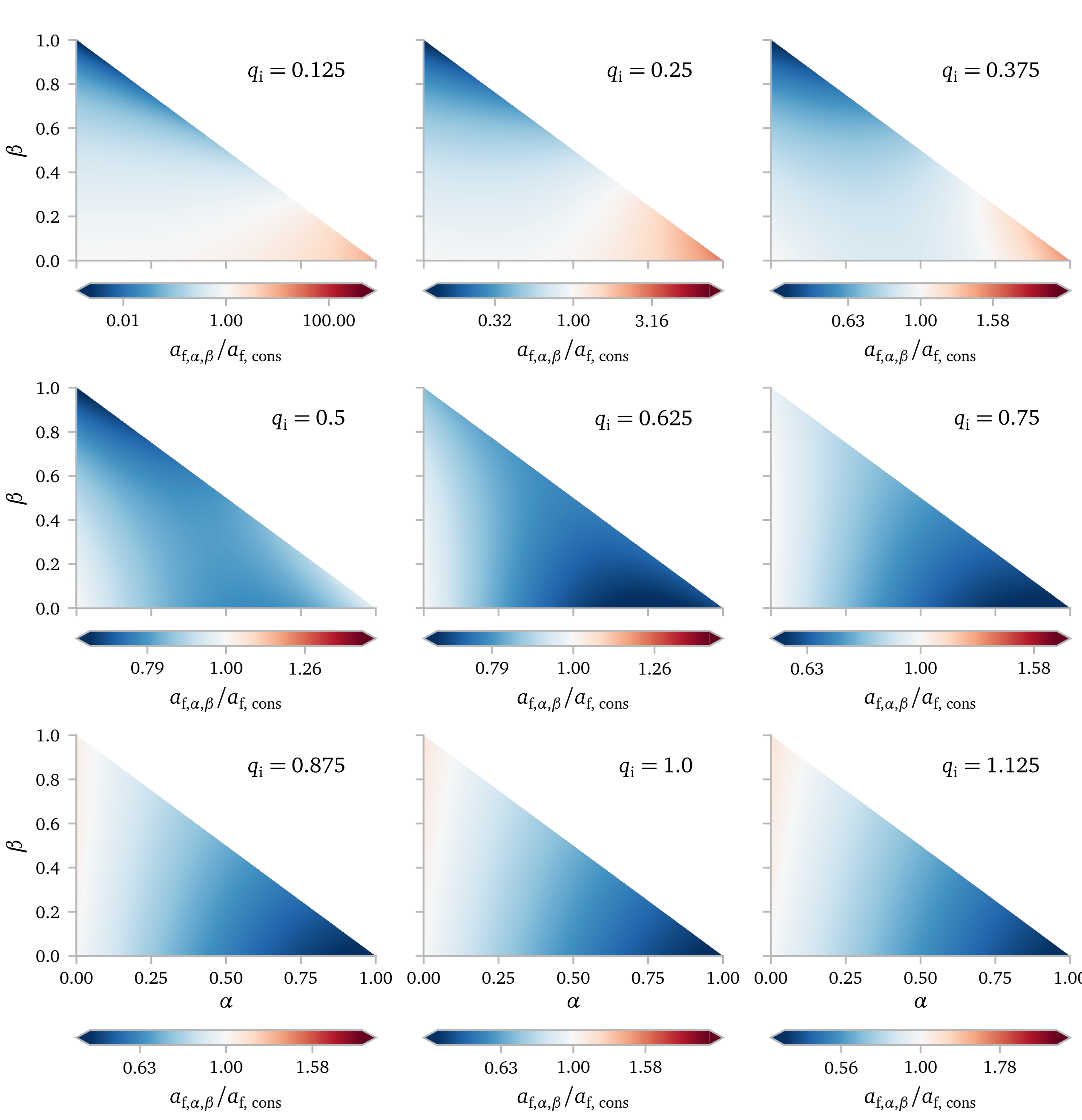
Interestingly, [1] nearly all systems, except for the initially very unequal ones with nearly no accretion, reach a final q that is greater than unity. I *think*, at this point, losing mass at the donor causes the system to lose more angular momentum than losing mass at the accretor.

1 But after a little bit of thought, very unsurprisingly!

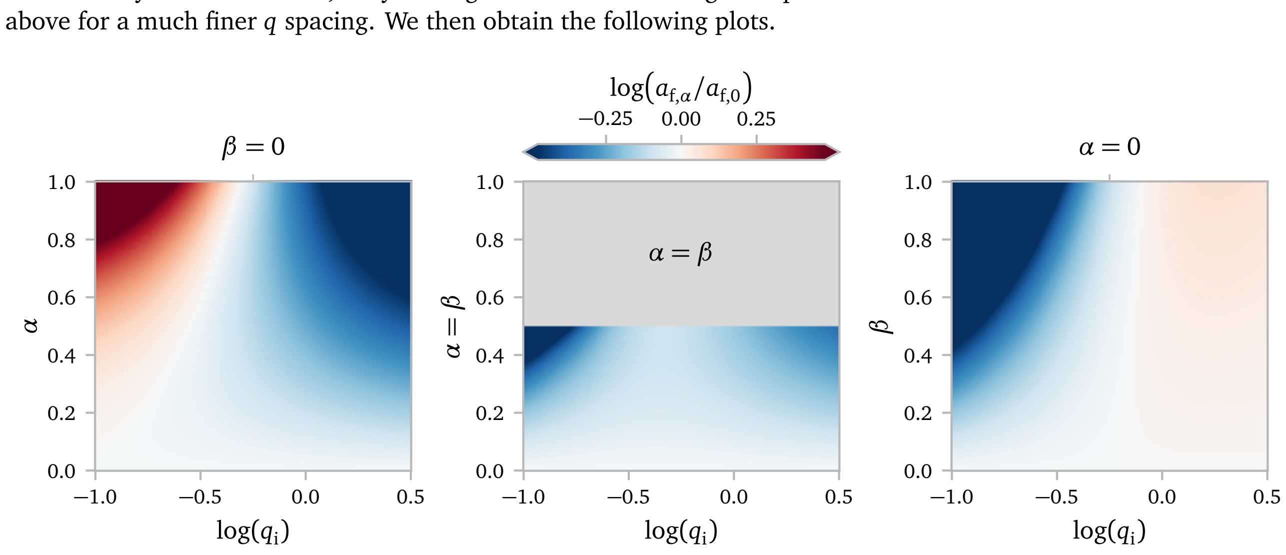
Below I plot the same figure, now for the average q during the evolution.



Now, we see that the systems where $q < 1$ align with the systems where the change in period for $\beta > 0$ is less than for $\beta = 0$.



Another way to visualize this, is by tracing cross sections through the plots above for a much finer q spacing. We then obtain the following plots.

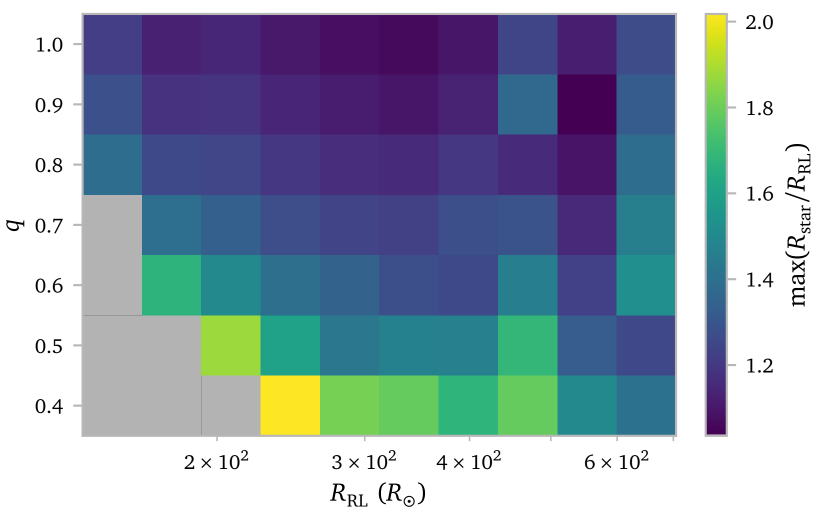


b. Grid

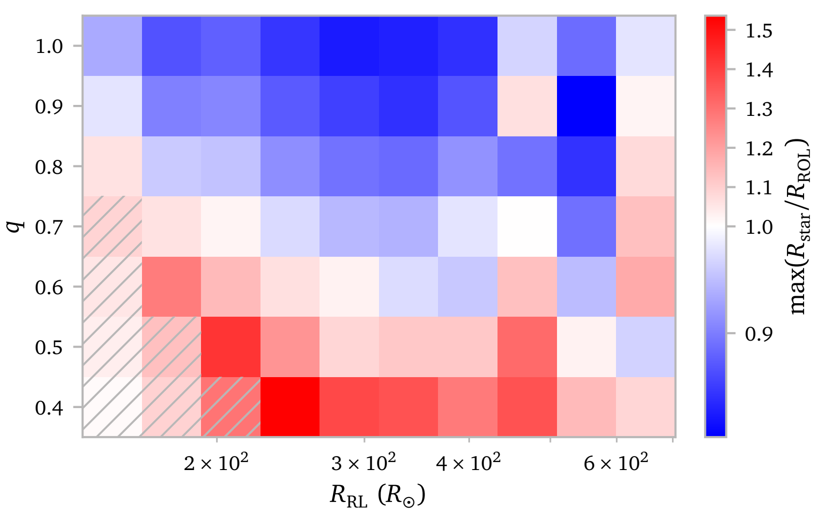
In this section, I analyze the grid that I simulated with starting Roche lobe Radii from 150 $R_\odot$ to 650 $R_\odot$ in 10 logarithmically spaced steps and 7 $q = M_\mathrm{a}/M_\mathrm{d}$ values from 0.4 to 1.

b.1 General properties

I start with some general properties of the models displayed in a 2d grid.

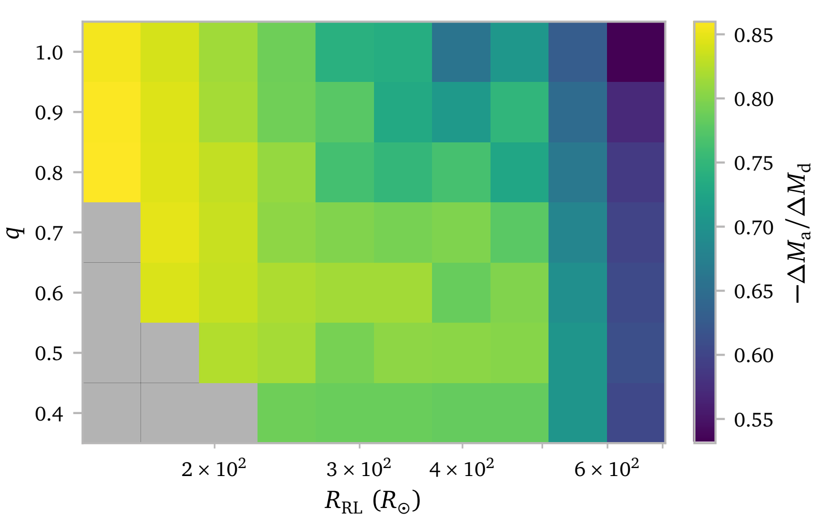


Here, I plot the maximum ratio of the star radius over the Roche lobe radius for every model. Initial q–values close to 1 seem to result in less extreme ratios. This makes sense, as for approximately conservative mass transfer, the orbit will reach its minimum separation when $q \approx 1$. For initial $q < 1$, this means the orbit will first shrink as the mass ratio approaches unity. This effect becomes less pronounced the closer the initial mass ratio is to unity. Once the winds become important, This relation becomes less clear. In the case of winds, the orbit will generally grow less for mass ratios close to unity than it will for mass ratios far from unity.

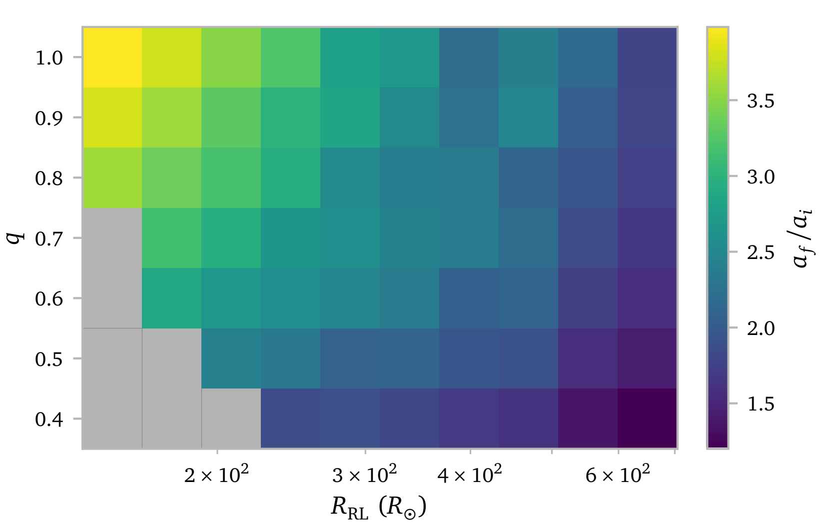


Similarly, we can plot the maximum ratio of the stellar radius over the volume equivalent outer Roche lobe radius using the analytical approximation from Temmink et al. (2023). The plot shows the same features but I use a divergent color map to show models where the ratio is greater and smaller than unity. Since the MESA models do not take outer lobe overflow into account, and the assumption of spherical symmetry can no longer be trusted when the stellar radius is larger than the outer Roche lobe, this is an important plot.

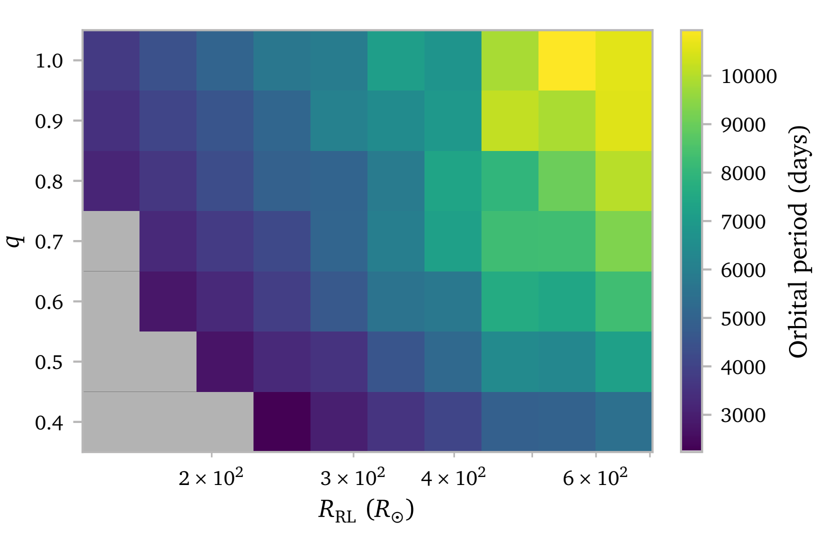
I think the third to last column has some surprising results. I investigate this in more detail in subsection b.2



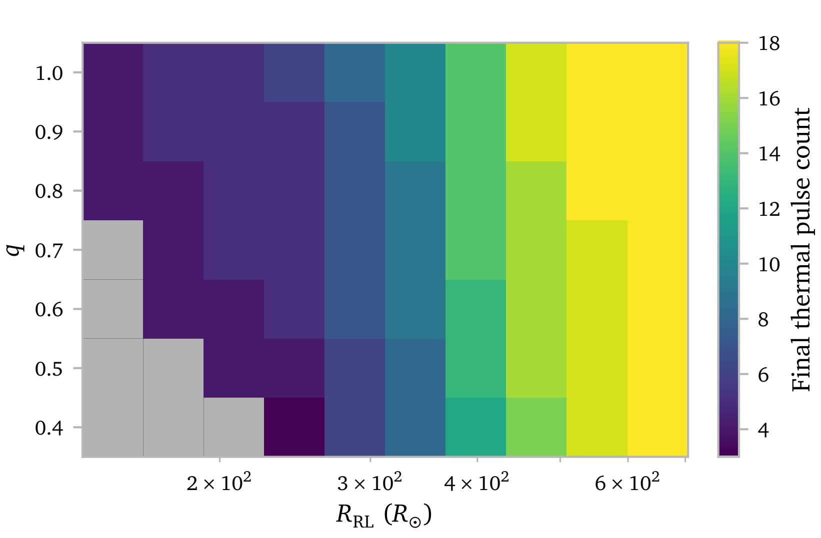
This plot shows the ratio of mass gained by the accretor and mass lost from the donor. A *higher* value means *more* of the mass that is lost from the donor successfully accretes on the accretor. The general trend of the plot seems to be: Larger initial Roche lobe radii lead to a smaller fraction of accreted mass. This makes sense since wind mass loss, which has a maximum accretion efficiency of about 30% becomes more important at larger radii.



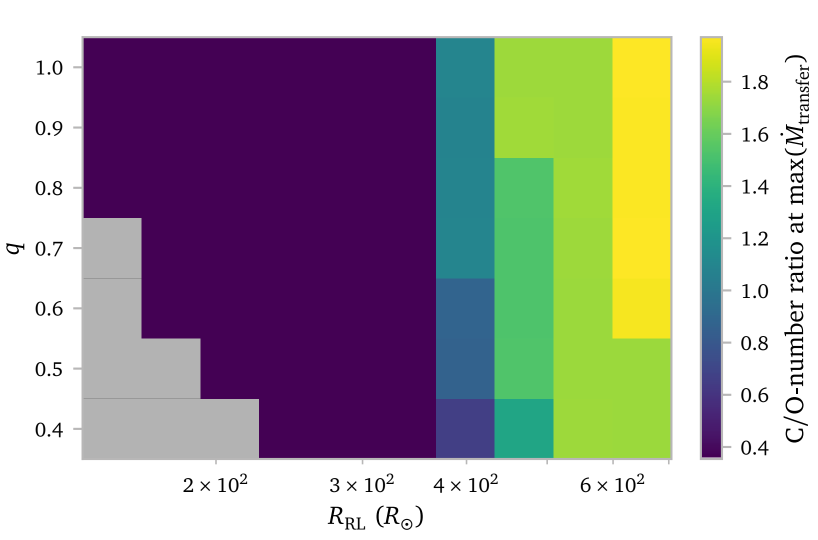
I think the results from this figure can be understood by combining the results section a. and the plot above. We see that the fraction of mass that is being accreted decreases with Roche lobe radius, and that the ratio of initial to final separation increases with q for conservative mass transfer. In general, a significant amount of mass is accreted, even when wind becomes imporatant, but when this is the case, the relation between q and $a_\mathrm{f}/a_\mathrm{i}$ becomes less pronounced.



This plot relates the amount of mass lost from the system, the plot above, and the initial periods. Systems with larger q tend to have larger initial separations. Systems with larger initial Roche lobe radii also tend to have larger initial separations. Combining this with mass lost due to winds, the largest periods occur for systems in the top right.



This the amount of thermal pulses the donor star experienced before becoming a post-AGB star. Perhaps the result is unsurprising. Stars starting with lower initial Roche lobe radii tend to lose their envelope quicker and thus experience less pulses. I guess the most interesting thing here is that stars with the same initial Roche lobe radius can experience a different number of thermal pulses depending on the mass of their companion.



This figure shows the number ratio of C/O at the time of the maximum mass transfer rate. Since most of the mass is transferred during a single mass transfer episode during a thermal pulse, this is a good first approximation for the C/O-ratio of the material that gets accreted onto the accretor. It follows the plot above nicely, which is very much expected. There is quite a sudden jump though in the C/O-ratio. I *think* this is quite typical of TPAGB stars (at least 2 $M_\odot$ TPAGB stars).

I think these plots, in particular the plot showing the C/O-ratio and the plot showing the final period, highlight that it is difficult to create the orbital properties of short period barium stars when assuming conservative RLOF mass transfer. Some mass probably has to be lost at the accretor to shrink the orbit sufficiently to achieve short period binaries.

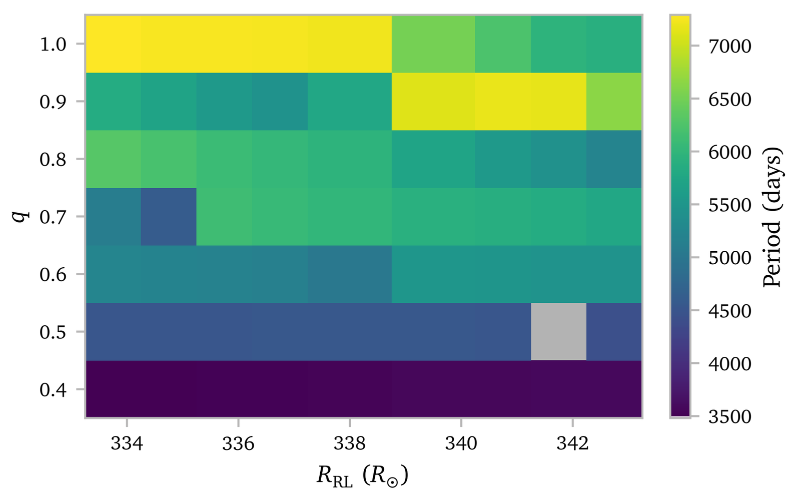
b.2 $R_{\mathrm{RL}} = 469.24\, R_{\odot}$

b.3 Fine Grid

In order to see how stable the results are to small variations in initial Roche lobe radius, I decided to simulate another grid. To test this, I decided to take a Roche lobe radius that I had already simulated and simulate 10 additional radii each differing by 1 $R_\odot$.

I chose the reference radius as $R_{\mathrm{RL}} \approx 338.75\ R_\odot$. This is because for this radius, the binary systems with different q interacted during three different pulses, with some systems barely avoiding significant mass transfer. The systems also interact nicely during the middle to end of the TPAGB, so wind can have some effects while not overpowering the evolution.

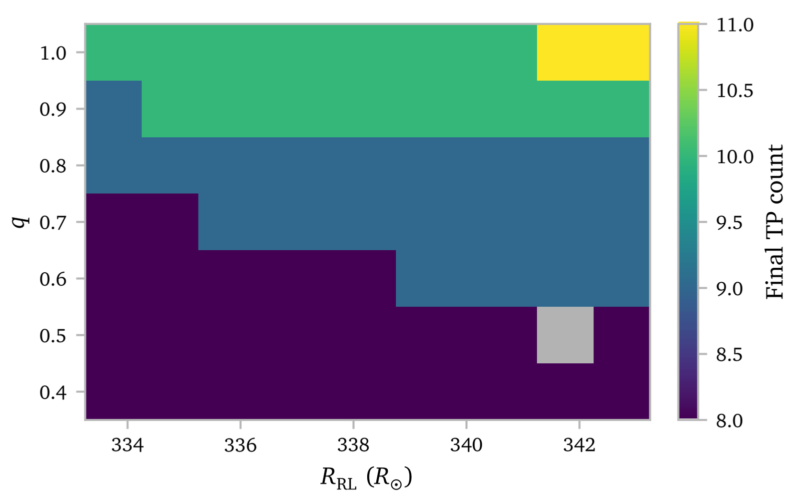
I was particularly interested in seeing how much the periods change with small perturbations in the initial Roche lobe radius. The plot below shows this.



In some cases, the differences are quite drastic. Nearly identical starting orbits can end with very different separations. I think it is important to try to understand where these differences come from.

Also, one of the models failed. It is interesting to look what caused this.

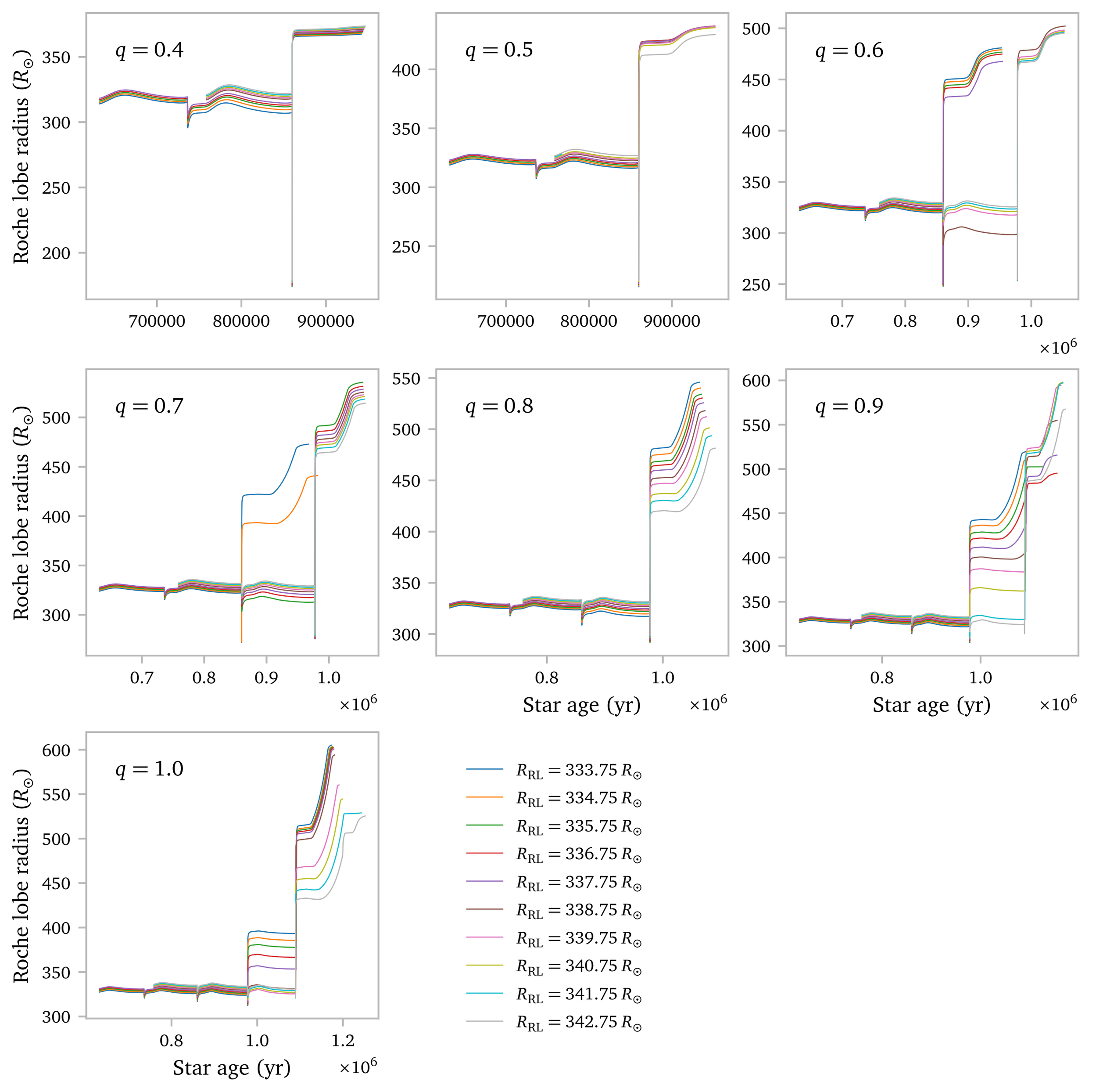
Presumably, this is due to interactions during a different thermal pulse. To check, we can look at the final TP count of the binary systems.



Although not a perfect match (for example, $q = 0.9$ shows nearly all systems experiencing the same number of pulses while clear differences in final periods exist) there does seem to be some relation between the thermal pulse count and the jumps in final period.

The plots below show the evolution of *all* models of this fine grid. They are split by q values. This makes it easier to compare. These are thus able to show the time evolution of the models of a single column in the plots above.

I start with a plot that shows the Roche lobe radius as a function of time.



The first two plots are not very interesting, all models seem to interact during the same pulse. The models stay order, meaning initially tighter orbits remain tighter after the evolution and overall, the differences between the models do not seem to diverge [2]. We see that the failing model in the $q = 0.5$ plot failed very early on in the evolution, without a clear reason.

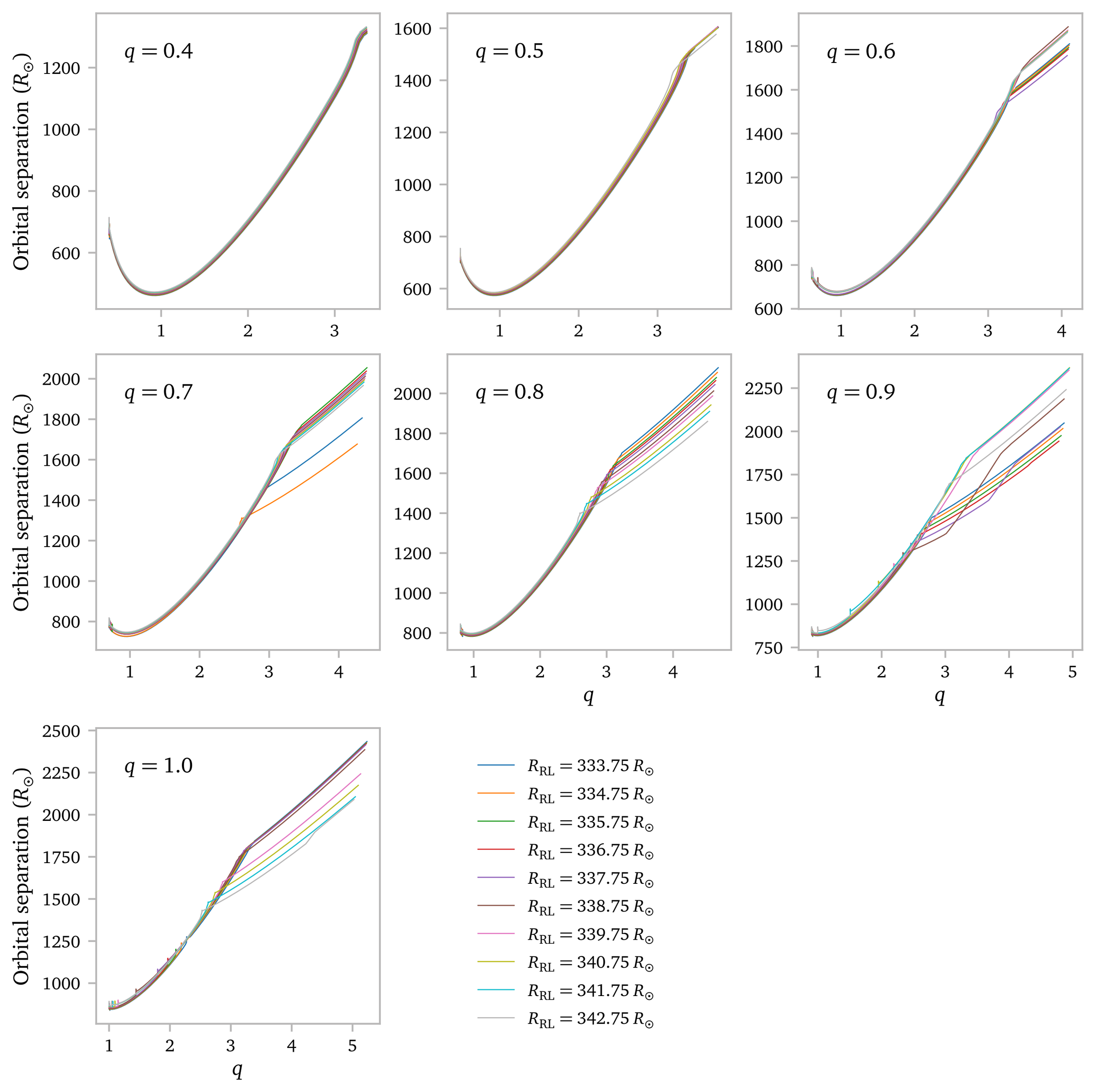
2 except for maybe the gray model in the $q = 0.5$ plot.

From $q = 0.6$ onwards [3], we see some clear divergence in behaviour for some of the models. The divergence arises due to the tighter binaries just reaching the required conditions for significant RLOF while the more extended ones barely miss these conditions.

3 except for the $q = 0.8$ models

Another behaviour starts to appear from $q = 0.6$ onwards. That is the reversal of the Roche lobe radii after mass transfer. While the order remains intact for the low q models, after $q = 0.6$, the initially tighter binaries end up at larger radii and the initially looser binaries. I really do not know what causes this.

Below i plot the evolution of the binary separation as a function of q.

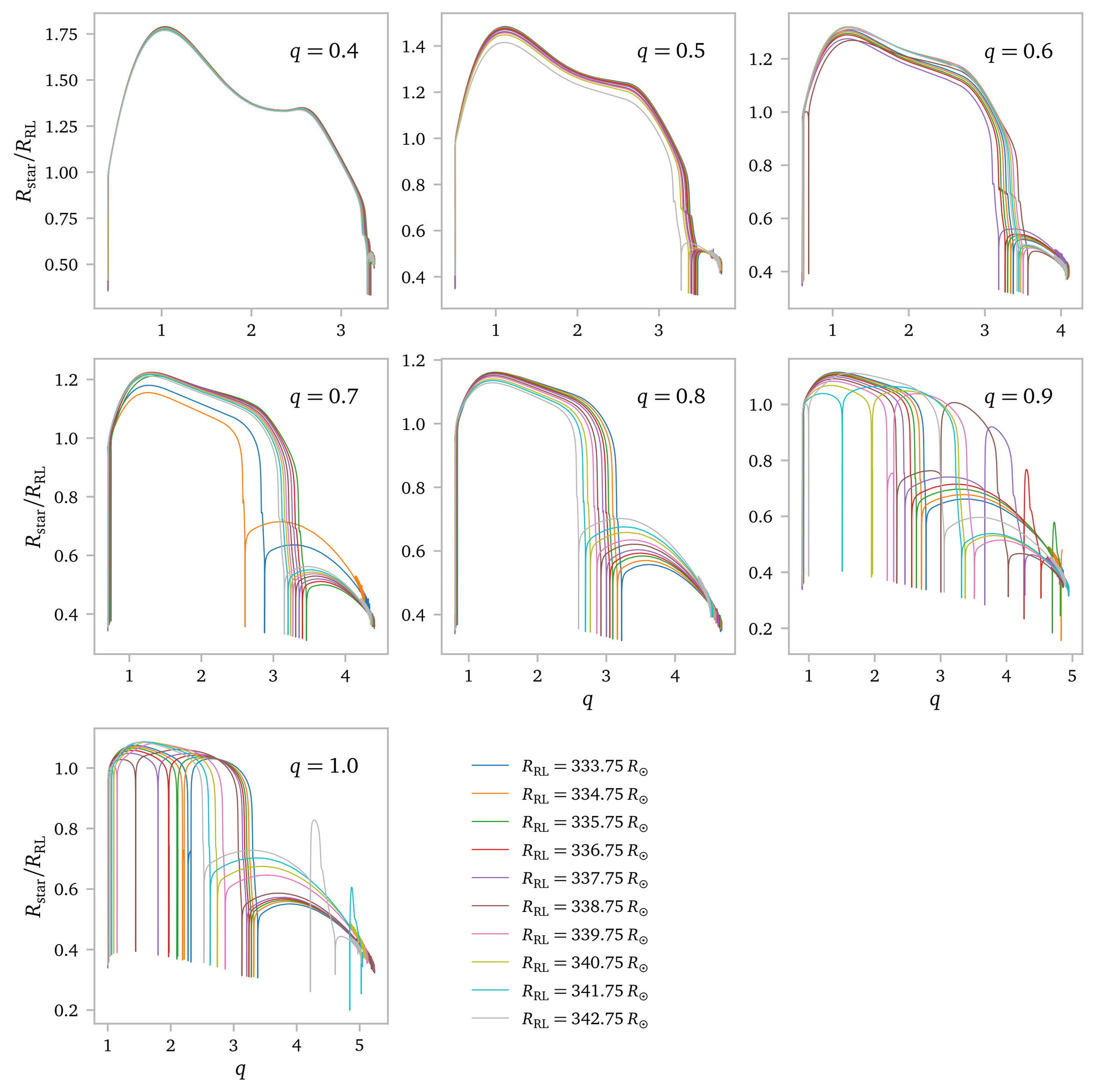


In this plot, it very easy to spot mass transfer due to winds[4] and the regions of RLOF. It is clear that for greater q, winds become more important. This must be because the smaller q models are able to sustain RLOF for longer before their orbits become too wide.

4 Straighter lines

Actually, for these radii, the cut-off for RLOF seems to lie at about $q = 3$. I am not entirely sure why this is and it probably has to do with the ability of the star to expand in response to mass loss.

The plot below shows the ratio of the star radius over the Roche lobe radius again as a function of the mass ratio.

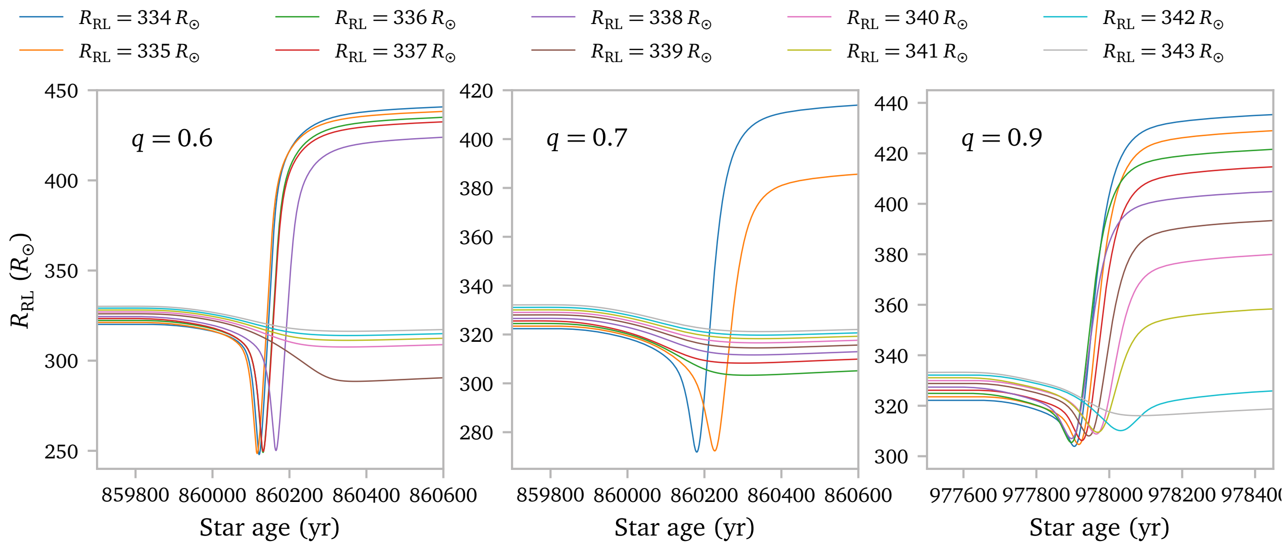


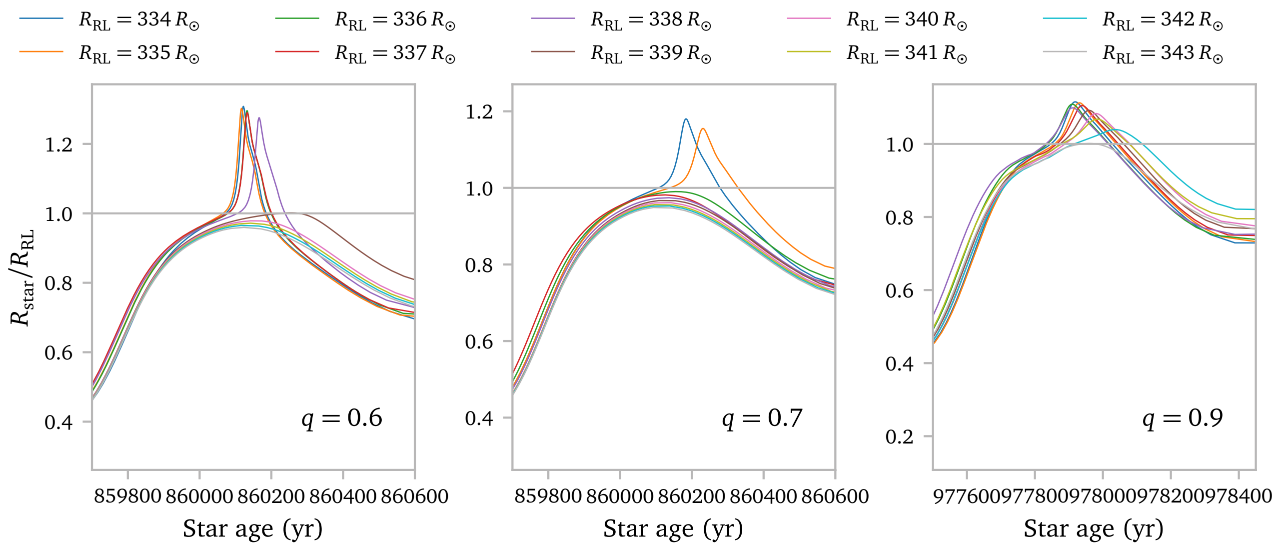
The same features are visible here. The last subplots get very messy, because the amount of mass transferred during the pulses is apparently very variable in these cases.

Lastly, we can zoom in a bit so hopefully see the separation between models that just barely start a RLOF mass transfer episodes and models that barely miss it. I look at the $q = 0.6, q = 0.7$ and $q = 0.9$ models because as seen in the figures above, there are clear separations visible here.

The upper panels show the Roche lobe radius, while the lower panels show the ratio. Unsurprisingly[5] , we see that models that fail to reach $R_{\rm star}/R_{\rm RL} > 1$ do not start their mass trasnfer episode. In this case, the Roche lobe radius shrinks in response to a narrow avoidance. Obviously, in response to the mass transfer episode, the Roche lobe radius increases significantly and a clear divide originates.

5 But still kind of cool to see!





References

Soberman, G. E., Phinney, E. S., & van den Heuvel, E. P. J. (1997) *Stability criteria for mass transfer in binary stellar evolution.*. A&A327, 620.

Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) *Coping with loss. Stability of mass transfer from post-main-sequence donor stars*. A&A669, A45.